How far the slice argument reaches

A characterisation of the criteria for which the efficient set of a multi-objective integer program is cheaply enumerable

Abstract

Let $D \subseteq \mathbb{Z}^n_+$ be finite and let $Z_1, \ldots, Z_p$ be criteria on D. If v is a non-dominated criterion vector then *every* $x \in D$ with $Z(x) = v$ is efficient, so the efficient set is exactly the union of the *slices* of the non-dominated set and no efficiency test is needed on any slice point. This argument assumes nothing about the criteria. What does need structure is the enumeration of a slice, and this note determines exactly how much.

We prove: (i) a criterion has one-hyperplane level sets with data depending affinely on the level *if and only if* it is linear fractional, on the set where the relevant denominator does not vanish — so within that shape the fractional case is maximal; (ii) the enumeration in fact needs only that each level set be a finite union of *polyhedra*, a strictly larger class that contains $|c^\top x + a|/(d^\top x + b)$ and maxima and minima of ratios; and (iii) a separate and strictly tighter condition governs the enumeration of the non-dominated set itself, namely that sublevel sets be half-spaces. The two reaches do not coincide, and the smaller one is what confines such a method to linear fractional criteria. We close with what the characterisation implies about neighbouring problem classes, and with a precise statement of what is *not* proved here.

1 Setting

Throughout, $D \subseteq \mathbb{Z}^n_+$ is a non-empty finite set — in the application $D = \{x \in \mathbb{Z}^n_+ : Ax \leqslant b\}$ with the system bounded — and $Z = (Z_1, \ldots, Z_p)$ is a vector of real-valued criteria on D, all to be maximised.

Definition 1. $y \in D$ *dominates* $x \in D$ when $Z(y) \geqslant Z(x)$ componentwise with at least one strict inequality. A point is *efficient* when nothing dominates it; $E(P_D)$ denotes the set of efficient points. A vector $v \in \mathbb{R}^p$ is *non-dominated* when $v = Z(x)$ for some $x \in D$ and no $Z(y)$ dominates it in the same sense; F denotes the set of non-dominated vectors.

Definition 2. For $v \in \mathbb{R}^p$ the *slice* at v is $S(v) = \{x \in D : Z(x) = v\}$.

2 The slice argument, which assumes nothing

Theorem 3. *If v is non-dominated then every $x \in S(v)$ is efficient.*

Proof. Let $x \in S(v)$ and suppose some $y \in D$ dominates x. Then $Z(y) \geqslant Z(x) = v$ with at least one strict inequality, so $Z(y)$ dominates v, contradicting the assumption that v is non-dominated. Hence no y dominates x. □

Corollary 4. $E(P_D) = \bigcup_{v \in F} S(v)$, *and the union is disjoint.*

Proof. ($\supseteq$) is Theorem 3. For ($\subseteq$), let x be efficient and put $v = Z(x)$. If some $Z(y)$ dominated v then y would dominate x, so $v \in F$ and $x \in S(v)$. Disjointness holds because $S(v) \cap S(w) = \emptyset$ for $v \neq w$. □

Remark 5. Theorem 3 uses no property of Z whatsoever — not linearity, not continuity, not finiteness of D beyond what makes the notions well posed. So the *characterisation* of $E(P_D)$ as a union of slices is available for arbitrary criteria. Everything that follows is about the cost of *enumerating* $S(v)$, which is a different question and the one where structure enters.

Remark 6. The practical content of Corollary 4 is that **no efficiency test is paid on any slice point**. A direct construction of $E(P_D)$ tests each candidate against the others; here the test is spent only in producing F, and the points are recovered by enumeration.

3 Where one hyperplane is exactly the fractional case

Call a criterion *linear fractional* when $Z(x) = (c^\top x + a)/(d^\top x + b)$ for fixed $c, d \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$.

Proposition 7. *If Z is linear fractional and $d^\top x + b \neq 0$ on D, then for every level v,*

$$S(v) = \{x \in D : (c - vd)^\top x = vb - a\},$$

an intersection of D with one hyperplane whose data is affine in v.

Proof. $Z(x) = v$ iff $c^\top x + a = v(d^\top x + b)$, since multiplying by the non-vanishing $d^\top x + b$ is reversible; rearranging gives the stated equation, whose coefficient vector $c - vd$ and right-hand side $vb - a$ are affine in v. □

The converse is the part that makes this a characterisation, and it holds for *any* affine dependence on the level rather than only for the parametrisation just produced.

Theorem 8 (Limit)**.** *Suppose there are $a_0, a_1 \in \mathbb{R}^n$ and $b_0, b_1 \in \mathbb{R}$ with*

$$\{x : Z(x) = v\} = \{x : (a_0 + va_1)^\top x = b_0 + vb_1\} \qquad \text{for every level } v.$$

Then on $\{x : b_1 - a_1^\top x \neq 0\}$ the criterion is linear fractional:

$$Z(x) \;=\; \frac{a_0^\top x - b_0}{b_1 - a_1^\top x}.$$

Together with Proposition 7: ***on the set where the denominator does not vanish, a criterion has one-hyperplane level sets with data affine in the level if and only if it is linear fractional.***

Proof. Fix such an x and put $v = Z(x)$. Then x lies in its own level set, so $(a_0 + va_1)^\top x = b_0 + vb_1$, that is $a_0^\top x - b_0 = v\,(b_1 - a_1^\top x)$. Dividing by the non-zero $b_1 - a_1^\top x$ gives the claim, with $c = a_0$, $a = -b_0$, $d = -a_1$ and $b = b_1$. □

Remark 9. The non-vanishing hypothesis is not a technicality to be dropped. It is the positive-denominator assumption under which the fractional problem is normally posed, restated; without it Z is not even continuous and the level-set description can fail. The theorem is accordingly stated *on* that set.

Remark 10. Theorem 8 is a *limitation*, and that is its value. It says there is nothing larger *of that shape* to look for: a method whose slices are single hyperplanes with affine data is a method for linear fractional criteria, whatever else one might hope.

4 But one hyperplane was never required

What an enumerator of $S(v)$ actually consumes is a system of linear constraints. Nothing forces that system to describe a single hyperplane, and nothing forces it to be a single system.

Proposition 11 (Reach, affine version)**.** *Suppose that for each level v there are finitely many affine subspaces $A_1(v), \dots, A_m(v)$, computable from v, with $\{x : Z(x) = v\} = \bigcup_i A_i(v)$. Then $S(v) = \bigcup_i \big(D \cap A_i(v)\big)$ is enumerable at m times the cost of the one-subspace case.*

Proof. Immediate: enumerate $D \cap A_i(v)$ for each i and take the union, discarding repetitions where the $A_i(v)$ meet. □

Example 12. $Z(x) = |c^\top x + a|/(d^\top x + b)$ with $d^\top x + b > 0$ is not linear fractional, and

$$\{x : Z(x) = v\} = \{x : c^\top x + a = v(d^\top x + b)\} \ \cup \ \{x : c^\top x + a = -v(d^\top x + b)\},$$

two hyperplanes. By Proposition 11 its slices are enumerable.

The same argument runs once more, and the second step is the one that reaches a criterion type of independent interest.

Proposition 13 (Reach, polyhedral version)**.** *Suppose that for each level v there are finitely many polyhedra $P_1(v), \dots, P_m(v)$, computable from v, with $\{x : Z(x) = v\} = \bigcup_i P_i(v)$. Then $S(v)$ is enumerable at m times the cost of one polyhedron.*

Proof. As before; an enumerator that accepts linear equations accepts linear inequalities at no structural cost, since both are linear constraints on the same variables. □

Example 14 (Minimax criteria)**.** Let $Z(x) = \max_{i \leqslant m} Z_i(x)$ with each Z_i linear fractional of positive denominator. Then

$$\{x : Z(x) = v\} \ = \ \bigcup_{j=1}^{m} \Big\{ x : Z_j(x) = v \ \text{ and } \ Z_i(x) \leqslant v \ \ (i \neq j) \Big\},$$

and each branch is one linear equation together with $m - 1$ linear inequalities — a polyhedron, not an affine subspace. Minimisation is the same with the inequalities reversed.

Remark 15. Propositions 11 and 13 are one observation applied twice: what a slice enumeration needs is not a *shape* but *finitely many linear systems*. What it cannot absorb is curvature.

5 The other half of such a method reaches less

Corollary 4 presupposes F. Producing F is a separate problem, and a method that searches criterion space by splitting around a visited point faces a different requirement.

Such a search, at a centre $\bar{x}$, divides the remaining region into $\{x : Z_k(x) > Z_k(\bar{x})\}$ and $\{x : Z_k(x) \leqslant Z_k(\bar{x})\}$. For each part to be a single linear program, both must be half-spaces.

Proposition 16 (Necessary condition for the split)**.** *If the split above is to express each part as one linear program for every centre, then every sublevel set $\{x : Z(x) \leqslant t\}$ must be a half-space.*

Proof. Taking $t = Z(\bar{x})$ as $\bar{x}$ ranges over D, the sublevel set is one of the two parts, hence a half-space; the other part is its complement, which is then a half-space as well. □

Remark 17 (What is not claimed)**.** Proposition 16 is a *necessary* condition, not a characterisation. A criterion could satisfy it and still not be usable, because a method must also *construct* those half-spaces from the problem data, and nothing above provides that in general. Sufficiency is available for linear fractional criteria, where after scaling the data to integers the quantity

$$e_k(x;\bar{x}) \;=\; D_k(\bar{x})\,D_k(x)\,\big(Z_k(x) - Z_k(\bar{x})\big)$$

is linear in x and integer-valued, so that $Z_k(x) > Z_k(\bar{x}) \iff e_k(x;\bar{x}) \geqslant 1$ exactly and both parts are explicit. An earlier draft of this material asserted an equivalence here; that assertion is withdrawn.

Proposition 18. *The condition of Proposition 16 is strictly stronger than that of Proposition 13.*

Proof. A half-space sublevel set has a hyperplane boundary, which is a level set, so a criterion satisfying Proposition 16 satisfies Proposition 11 with $m = 1$, hence Proposition 13. Strictness: in Example 12 the sublevel set $\{|c^\top x + a| \leqslant v(d^\top x + b)\}$ is the intersection of two half-spaces and its complement is not convex; in Example 14 it is $\bigcap_i \{Z_i \leqslant v\}$, a polyhedron whose complement is a union. Neither is a half-space, while both have enumerable slices. □

Remark 19. Proposition 18 explains a feature of such methods that is otherwise puzzling. They are described as methods for linear fractional programs, yet their completion step is indifferent to the criteria. The reason is that the two halves have different reaches, and **the smaller one names the method**. For a criterion between the two — Example 14 is one — the efficient set can be completed exactly once F is known, but F must be obtained some other way.

6 A consequence for the decision function

Theorem 20 (Admissible decision functions)**.** *Let $E(P_D)$ be known. Then for any function f defined on D, both* $\max\{f(x) : x \in E(P_D)\}$ *and* $\operatorname{argmax}\{f(x) : x \in E(P_D)\}$ *are obtained by evaluating f on $E(P_D)$. No linearity, monotonicity, continuity, differentiability or ratio form is required of f. The same holds for the non-dominated subset of $E(P_D)$ under finitely many such functions.*

Proof. $E(P_D)$ is finite. □

Remark 21. The proof is the whole of the content, and stating the theorem would be idle were it not for Corollary 4 and Remark 6: the point is not that maximising over a finite set is easy, but that the slice argument makes $E(P_D)$ affordable, and that this is what the affordability buys. Methods that never construct $E(P_D)$ must instead derive their cuts and bounds from the form of f, and so are confined to the forms they can differentiate or linearise.

7 What the characterisation says about neighbouring classes

Theorem 8 and Proposition 13 classify a criterion by inspection of its level sets.

Example 22 (Sums of a linear and a fractional term)**.** Let $Z(x) = c^\top x + (p^\top x + \alpha)/(q^\top x + \beta)$. Clearing the denominator at level v,

$$(c^\top x)(q^\top x + \beta) + p^\top x + \alpha - v\,(q^\top x + \beta) \;=\; 0,$$

whose quadratic part is $(c^\top x)(q^\top x)$, with matrix $(cq^\top + qc^\top)/2$. That matrix is non-zero whenever c and q both are, so the level set is a conic. A conic is not a finite union of polyhedra, so Proposition 13 does not apply and the slice machinery does not reach this class.

The same computation disposes of criteria that are products, and of quadratic criteria, for the same reason: curvature. Example 22 is worth isolating because the class it describes is studied in its own right, so the characterisation says in advance that a slice-based method and a method for that class are addressing disjoint problems.

8 What is not proved here

1. **Proposition 16 is not a characterisation.** See Remark 17. Sufficiency is shown only for linear fractional criteria.

2. **Nothing here bounds the cost.** Propositions 11 and 13 say a slice is enumerable, not that it is small. The size of $S(v)$ and the number of branches are properties of the instance.

3. **Theorem 8 concerns level sets with data affine in the level.** A criterion whose level sets are single hyperplanes with data depending on v in some other way is not covered, and we do not claim such criteria cannot exist.

4. **The classification of a published problem class** — as in Example 22 — rests on the class as stated. The algebra is ours; the identification of which class a given paper treats is not a mathematical claim.